

Assignment_7

SalesLT.Customer.txt

Assignment 7: ADF Pipeline:

Assignment 7: ADF Pipeline 3

Easy Questions:

1. Create a pipeline named 'GetMetaData_Example' to fetch the list of all the files available inside the ADLS folder.
2. Create a pipeline name 'Foreach_Example' to copy the Customer & Customer address table data into ADLS using the one copy activity.

Tough Questions:

3. Create a pipeline that first checks whether the Customer.csv file is available in the folder location. If it is available it will copy the file data to the Customer Table.
4. Create a pipeline that will create the incremental pipeline using the table instead of HighWaterMark File.
5. Create a pipeline name 'Foreach_Example2' to copy the below information using just one copy activity.
 - All the product table data where productid > 100
 - All the customer table data where Customer id>100 and Customer id <1000

.....

Solution 1:

- 1. Create a pipeline named 'GetMetaData_Example' to fetch the list of all the files available inside the ADLS folder.**

 Publish all 3

Ass_7_1_Get_Meta... ADLS_Generic_CSV_DS ADLS_Customer_CSV...

Activities

met

General

 Get Metadata

 Validate  Debug  Add trigger

Get Metadata



Fetch the list of all the files



General Settings User properties

Name *

Fetch the list of all the files

[Learn more](#)

Description

Create a pipeline named
GetMetadata_Example to fetch the
list of all the files available inside the

Activity state ^①

☒ Activated

☐ Deactivated

Timeout ^①

0.12:00:00

Retry ^①

0

Retry interval (sec) ^①

30

ADLS_Generic_CSV_DS ADLS_Customer_CSV...

✓ Validate ▶ Debug ⚡ Add trigger

Get Metadata ✓
Fetch the list of all the files
🗑️ </> 📄 ➡

General Settings User properties

Dataset *

ADLS_Customer_CSV_Dataset ▼ Open + New Learn more

Field list *

+ New | 🗑️ Delete

☐ Argument

☐ Exists ▼

Filter by last modified ⓘ

Start time (UTC)

End time (UTC)

Skip line count

 Publish all 

 Ass_7_1_Get_Meta...  ADLS_Generic_CSV_DS  ADLS_Customer_CS... 



DelimitedText
ADLS_Customer_CSV_Dataset

Connection Schema Parameters

Linked service *

 ADLS_Azure_DataLakeStorage1_Link...  Test connection  Edit  New [Learn more](#) 

File path *

landing / CSV / SalesLT.Customer.txt  Browse | 

Compression type

Select... 

Column delimiter 

Comma (,) 

Row delimiter 

Default (\r,\n, or \r\n) 

Encoding 

Default(UTF-8) 

Quote character 

Double quote (") 

Escape character 

Backslash (\) 

First row as header 



Null value 

✓ Validate ▶ Debug ⚡ Add trigger

Get Metadata

✓

✗

✓

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Fetch the list of all the files

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→

General **Settings** User properties

Dataset *

 ADLS_Customer_CSV_Dataset ▼

 Open  New [Learn more](#) 

Field list *

 New |  Delete

☐ Argument

☐ Exists ▼

Filter by last modified ⓘ

Start time (UTC)

End time (UTC)

Skip line count

● ADLS_Generic_CSV_DS ● ADLS_Customer_CSV...

⌵ ⌵

✓ Validate ▶ Debug ⚡ Add trigger

Get Metadata

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Fetch the list of all the files

✓

Parameters Variables Settings Output

Pipeline run ID: 805bed0a-99fc-4851-8b95-9e94982226bd [📷] 🔄 ⓘ

All status ▾

Showing 1 - 1 of 1 items

Activity name ⬆️⬆️	Activity status ⬆️⬆️	Activity type ⬆️⬆️	Run status
Fetch the list of all the files	✓ Succeeded	Get Metadata	2/13/20

Get Metadata

 Fetch the list of all the files 

Output

 Copy to clipboard

```
{
  "exists": true,
  "effectiveIntegrationRuntime":
"AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 0
  },
  "billingReference": {
```

 (

type ↑

Fetch the list of all the files  Succeeded Get Metadata

TO find all files in a folder

ADLS_Generic_CSV_DS

✓ Validate

▶ Debug

⚡ Add trigger

Trigger test runs of the current pipeline without publishing your changes to the service

Get Metadata

Fetch the list of all the files

🗑️

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📄

➔

General

Settings

User properties

Name *

Fetch the list of all the files

[Learn more](#)

Description

Create a pipeline named GetMetadata_Example to fetch the list of all the files available inside the

Activity state ⓘ

☒ Activated ☐ Deactivated

Timeout ⓘ

0.12:00:00

Retry ⓘ

0

Retry interval (sec) ⓘ

30

Secure output ⓘ

☐

ADLS_Generic_CSV_DS

✓ Validate ▶ Debug ⚡ Add trigger

Get Metadata

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Fetch the list of all the files

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✓

✗

➔

General **Settings** User properties

Dataset *

ADLS_Generic_CSV_DS

[Open](#) [+ New](#) [Learn more](#)

Dataset properties ⓘ

Name	Value
------	-------

folderName_param	CSV
------------------	-----

Field list * [+ New](#)

	Start time (UTC)	End time (UTC)
Filter by last modified ⓘ		

Skip line count	
-----------------	--

mission00Gopio1012024adl

Search factory and documentation

Public preview of Microsoft Fabric, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Validate all [Publish all](#)

Ass_7_1_Get_Meta...

ADLS_Generic_CSV_...

19

Customer_J...

3_2

3_2

DD

sample

n_Product_...

eline_2

n_Product_...

sample

shold_file

Folder

_CSV

Pipe

I

SV

DB_ADLS

DB_ADLS_F...

>N

e_Pipeline

view

0

13

Asset

 DelimitedText
ADLS_Generic_CSV_DS

Connection

Schema

Parameters

Linked service *

File path *

Compression type

Column delimiter

Row delimiter

Encoding

Quote character

Escape character

First row as header

Null value

ADLS_Azure_DataLakeStorage1_Link...

Test connection

Edit

New

Learn more

landing

@dataset().folderName_param

File name

Browse

Prev

Select...

Comma (,)

Default (r,n, or v,n)

Default(UTF-8)

Double quote (")

Backslash (\)

☒

ure

Upgrade

Search resources, services, and docs (G+/)

[unts](#) > [gopio1012024adls](#) | Containers >

...

Upload

Add Directory

Refresh

Rename

Delete

Change tier

Acquire lease

Break lease

Give feedback

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / CSV

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ [.]		
☐ SalesLT.Customer.txt	2/4/2024, 4:23:03 PM	Hot (Inferred)
☐ SalesLT.Product.txt	1/28/2024, 10:12:56 PM	Hot (Inferred)

List all files

Meta...

ADLS_Generic_CSV_DS

Validate

Debug

Add trigger

ita

Get Metadata

Fetch the list of all the files

General

Settings

User properties

Dataset *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name

Value

folderName_param

CSV

Field list *

New

Delete

Argument

Child items

Filter by last modified

Start time (UTC)

Child items

End time (UTC)

Input

↗ ✕

📄 Copy to clipboard

```
"dataset": {  
  "referenceName": "ADLS_Generic_CSV_DS",  
  "type": "DatasetReference",  
  "parameters": {  
    "folderName_param": "CSV"  
  }  
},  
"fieldList": [  
  "childItems"  
]
```

↻ ⓘ

type ↕

Fetch the list of all the files ✔ Succeeded Get Metadata

Input

↗ ✕

📄 Copy to clipboard

```
"fieldList": [  
  "childItems"  
],  
"storeSettings": {  
  "type": "AzureBlobFSReadSettings",  
  "enablePartitionDiscovery": false  
},  
"formatSettings": {  
  "type": "DelimitedTextReadSettings"  
}
```

↻ ⓘ

type

Fetch the list of all the files ✔ Succeeded Get Metadata

ADLS_Generic_CSV_DS

Validate

Debug

Add trigger

Get Metadata

Fetch the list of all the files

Output

Copy to clipboard

childItems

[

nameSalesLT.Customer.txt

typeFile

,

nameSalesLT.Product.txt

typeFile

]

Fetch the list of all the files

Succeeded

Get Metadata

ADLS_Generic_CSV_DS

Validate Debug Add trigger

Get Metadata

Fetch the list of all the files

Output

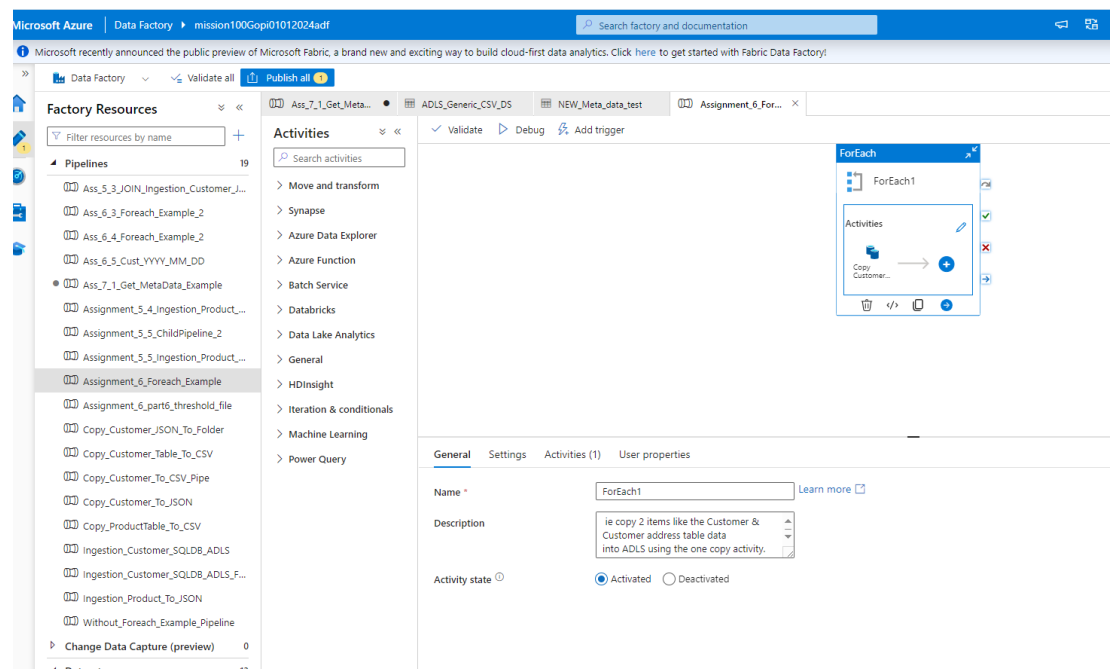
Copy to clipboard

```
{
  "childItems": [
    {
      "name": "SalesLT.Customer.txt",
      "type": "File"
    },
    {
      "name": "SalesLT.Product.txt",
      "type": "File"
    }
  ]
}
```

Fetch the list of all the files Succeeded Get Metadata 2/13/2024, 10:18

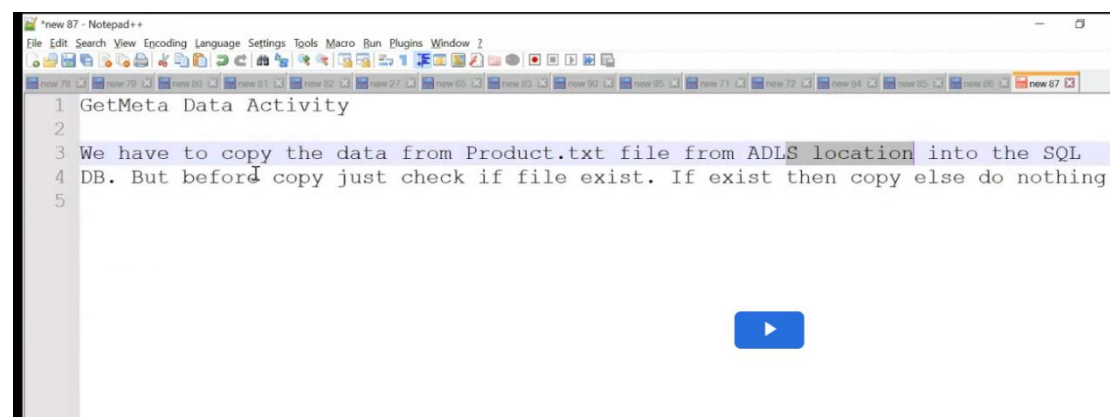
2. Create a pipeline name 'Foreach_Example' to copy the Customer & Customer address table data into ADLS using the one copy activity.

--- Solution Same as Assignment 6 part 2 (question 2) -- duplicate -- please refer it



Tough Questions:

3. Create a pipeline that first checks whether the Customer.csv file is available in the folder location. If it is available it will copy the file data to the Customer Table.



« Validate Debug Add trigger

Get Metadata

i

Fetch the list of all the files

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✓

✗

➔

General Settings User properties

Name *	Fetch the list of all the files	Learn more
Description	Create a pipeline named GetMetaData_Example to fetch the list of all the files available inside the	
Activity state ①	☒ Activated ☐ Deactivated	
Timeout ①	0.12:00:00	
Retry ①	0	
Retry interval (sec) ①	30	
Secure output ①	☐	
Secure input ①	☐	

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[Publish all](#) 3

Ass_7_1_Get_Meta... ADLS_Generic_CSV... NEW_Meta_data_test Assignment_6_Fore... Ass_7_ques_3



DelimitedText
ADLS_Generic_CSV_DS

Connection Schema Parameters

Linked service *	ADLS_Azure_DataLakeStorage1_Link... Test connection Edit New Learn more
File path *	landing / @dataset().folderName_param / SalesLT.Customer.txt Browse
Compression type	Select...
Column delimiter ①	Comma (,)
Row delimiter ①	Default (\r,\n, or \r\n)
Encoding ①	Default(UTF-8)
Quote character ①	Double quote (")
Escape character ①	Backslash (\)
First row as header ①	☒
Null value ①	

Get Metadata

i

Check if Customer CSV file exist

✓

✓

✗

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📄

→

General

Settings

User properties

Dataset *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name	Value
folderName_param	Customer

Field list *

New

Delete

Argument

Item name

Filter by last modified

Start time (UTC)

Item name

End time (UTC)

Skip line count

Get Metadata

Check if Customer CSV file exist

Input

Copy to clipboard

```
{
  "dataset": {
    "referenceName": "ADLS_Generic_CSV_DS",
    "type": "DatasetReference",
    "parameters": {
      "folderName_param": "Customer"
    }
  },
  "fieldList": [
```

Check if Customer CSV file ex...

Queued

Get Metadata

↗ ✕

Copy to clipboard

```
{
  "dataset": {
    "referenceName": "ADLS_Generic_CSV_DS",
    "type": "DatasetReference",
    "parameters": {
      "folderName_param": "Customer"
    }
  },
  "fieldList": [
```

Input

 Copy to clipboard

```
{
  "type": "DatasetReference",
  "parameters": {
    "folderName_param": "Customer"
  },
  "fieldList": [
    "itemName"
  ],
  "storeSettings": {
```

Check if Customer ...    Succeeded

Get M

Output

 Copy to clipboard

```
{
  "itemName": "SalesLT.Customer.txt",
  "effectiveIntegrationRuntime":
"AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 10
  },
  "billingReference": {
```

Check if Customer CSV file ex...  Succeeded

Get Metadata



Get Metadata



Check if Customer CSV file exist

Output



Copy to clipboard

```
{  
  "itemName": "SalesLT.Customer.txt",  
  "effectiveIntegrationRuntime":  
    "AutoResolveIntegrationRuntime (East US)",  
  "executionDuration": 0,  
  "durationInQueue": {  
    "integrationRuntimeQueue": 10  
  },  
  "billingReference": {
```



	type	Run start
Check if Customer ...		Succeeded
Get Metadata		2/13/2024, 10:47:

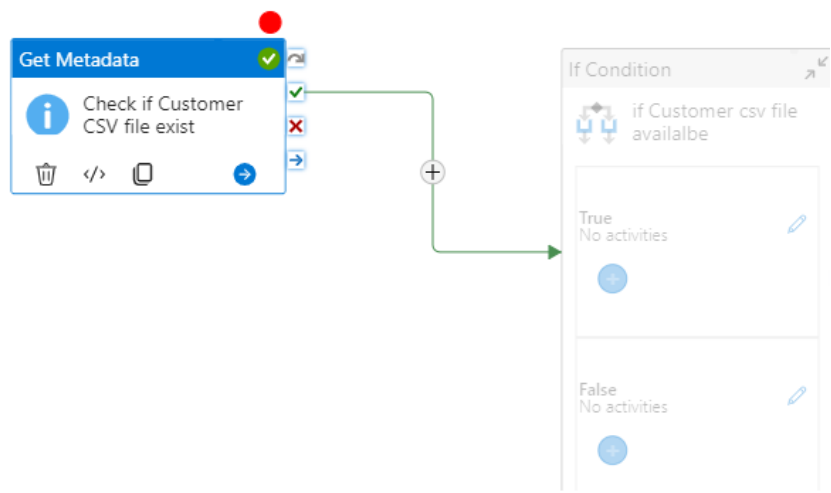




DelimitedText
ADLS_Generic_CSV_DS

Connection Schema Parameters

Linked service *	ADLS_Azure_DataLakeStorage1_Link...Test connection Edit + New Learn more		
File path *	landing	/	@dataset().folderName_paramSalesLT.Customer.txt
Compression type	Select... ▾		
Column delimiter ①	Comma (,)		
Row delimiter ①	Default (\r,\n, or \r\n)		
Encoding ①	Default(UTF-8)		
Quote character ①	Double quote (")		
Escape character ①	Backslash (\)		
First row as header ①	☒		
Null value ①			



General **Settings** User properties

Dataset * ADLS_Generic_CSV_DS Open New Learn more

Dataset properties ⓘ

Name	Value
folderName_param	Customer

Field list * New Delete

☐ Argument

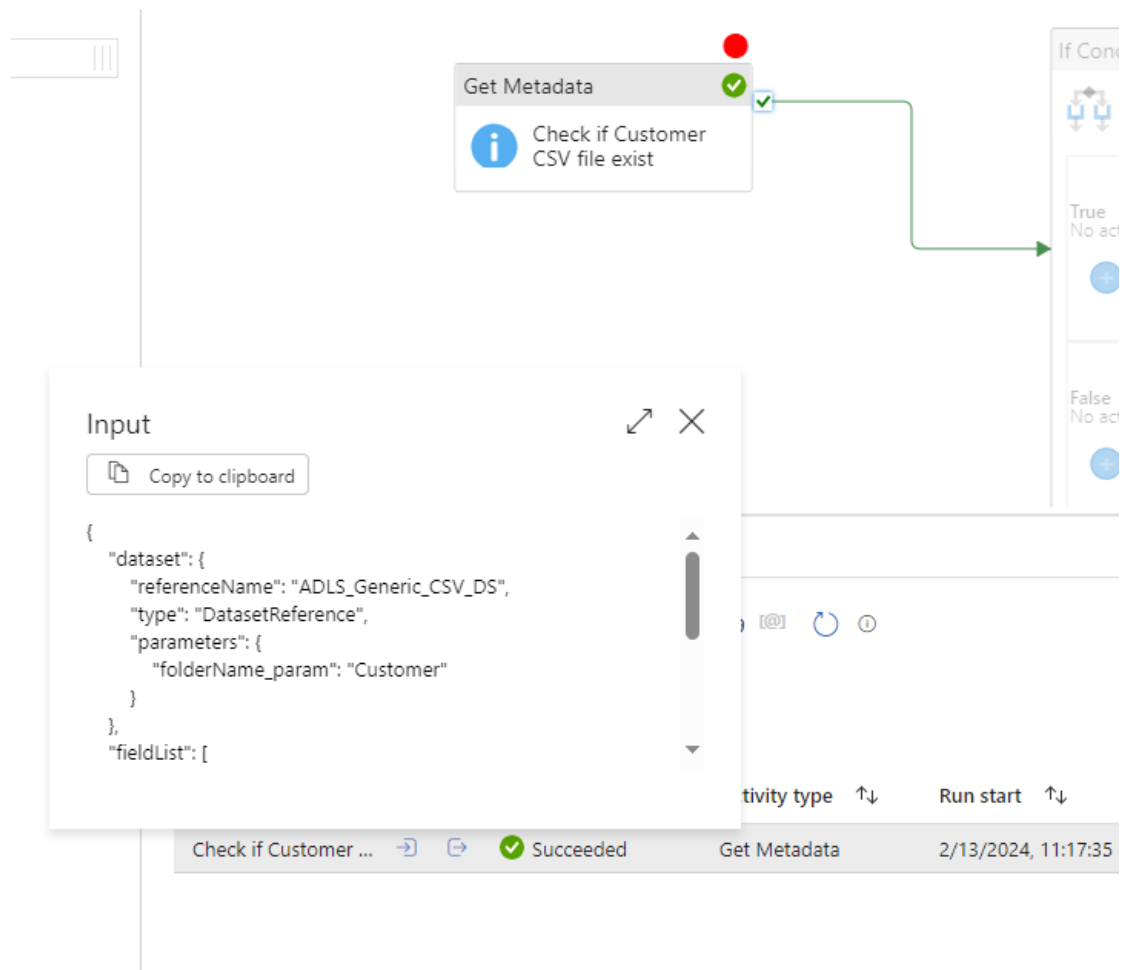
☐ Exists

Filter by last modified ⓘ

Start time (UTC)

End time (UTC)

Skip line count





Get Metadata



Check if Customer
CSV file exist



Output



Copy to clipboard

```
{
  "exists": true,
  "effectiveIntegrationRuntime":
"AutoResolveIntegrationRuntime (East US)",
  "executionDuration": 0,
  "durationInQueue": {
    "integrationRuntimeQueue": 0
  },
  "billingReference": {
```



type ↑

Check if Customer ...



Succeeded

Get Metadata

Get Metadata

Check if Customer CSV file exist

If Condition

if Customer csv file availalbe

True

No activities

False

No activities

General

Activities (0)¹

User properties

Expression ⓘ

@activity('Check if Customer CSV file...

Case	Activity
True	No activities
False	No activities

to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

eric_CSV_DS • NEW_Meta_data_test Assignment_6_Fore... Ass_7_ques_3

date ▶ Debug ⚡ Add trigger

Get Metadata

Check if Customer CSV file exist

If Condition

if Cust... availal

True

No activities

False

No activities

al

Activities (0)¹

User properties

sion ⓘ

@activity('Check if Customer CSV file...'

Activity
No activities
No activities

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variable

@activity('Check if Customer CSV file exist').output.exists

Clear contents

Activity outputs

Parameters

System variables

Functions

Variables

Search

Check if Customer CSV file exist

Check if Customer CSV file exist activity output

Check if Customer CSV file exist

Check if Customer CSV file exist pipeline return value

Check if Customer CSV file exist columnCount

Number of columns in the file or relational table

Check if Customer CSV file exist contentMD5

MD5 of the file. Applicable only to files

Check if Customer CSV file exist exists

Whether a file, folder, or table exists

Check if Customer CSV file exist itemName

Name of the file or folder

Check if Customer CSV file exist itemType

Type of the file or folder. Returned value is File or Folder

Check if Customer CSV file exist lastModified

new and exciting way to build cloud first data analytics. Click [here](#) to get started with Fabric Data Factory.

ADLS_Generic_CSV_DSNEW_Meta_data_testAssignment_6_Fore...Ass_7_ques_3

ValidateValidate copy runtimeDebugAdd trigger

Ass_7_ques_3 > if Customer csv file availalbe > True activities

Copy data

 Copy file data to Customer table

 </>  

GeneralSourceSinkMappingSettingsUser properties

Name *

Copy file data to Customer table

[Learn more](#)

Description

Copy Customer.csv file data to the Customer Table.

Activity state ①

☒ Activated ☐ Deactivated

Timeout ①

0.12:00:00

Retry ①

0

Retry interval (sec) ①

30

Secure output ①

☐

Ass_7_ques_3 > if Customer csv file availalbe > True activities

Copy data

Copy file data to Customer table

General **Source** Sink Mapping Settings User properties

Source dataset * ADLS_Generic_CSV_DS [Open](#) [New](#) [Preview data](#) [Learn more](#)

Dataset properties ⓘ

Name	Value
folderName_param	Customer

File path type ☒ File path in dataset ☐ Wildcard file path ☐ List of files ⓘ

Filter by last modified ⓘ

Start time (UTC) End time (UTC)

Recursively ⓘ ☒

Enable partitions discovery ⓘ ☐

[Publish all](#)

Ass_7_1_Get_Meta... ADLS_Generic_CSV_... NEW_Meta_data_test Assignment_6_Fore... Ass_7_ques_3 ADLS_Customer_CSV... SQL_Customer_Te



DelimitedText
ADLS_Generic_CSV_DS

Connection Schema Parameters

Linked service * ADLS_Azure_DataLakeStorage1_Link... [Test connection](#) [Edit](#) [New](#) [Learn more](#)
Connection successful

File path * / [Browse](#) [Preview data](#) [Detect](#)

Compression type

Column delimiter ⓘ

Row delimiter ⓘ

Encoding ⓘ

Quote character ⓘ

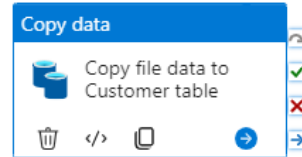
Escape character ⓘ

First row as header ⓘ ☒

Null value ⓘ

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_7_ques_3 > if Customer csv file availalbe > True activities



General Source Sink Mapping Settings User properties

Sink dataset *	Ass_7_SQL_Customer_Table_DS	Open	New	Learn more
Write behavior	☒ Insert ☐ Upsert ☐ Stored procedure			
Bulk insert table lock ①	☐ Yes ☒ No			
Table option	☒ None ☐ Auto create table ①			
Pre-copy script ①				
Write batch timeout ①	e.g. 00:30:00			
Write batch size ①				
Max concurrent connections ①				

Microsoft Store, or third-party software may require additional software or hardware to get started with Azure Data Factory.

Publish all 4

Ass_7_1_Get_Meta... ADLS_Generic_CSV_DS NEW_Meta_data_test Assignment_6_Fore... Ass_7_ques_3 ADLS_Cus...

SQL Azure SQL Database
Ass_7_SQL_Customer_Table_DS

Connection Schema Parameters

Linked service * SQL_DB_Linked_service1 Test connection Edit + New Learn more

Table SalesLT.Customer Refresh Preview data

☐ Enter manually

Error details
Copy file data to Customer table

Details
Failure happened on 'Sink' side.
ErrorCode=SqlOperationFailed, Type=Microsoft.DataTransfer.Common.Shared.HybridDeliveryException, Message=A database operation failed. Please search error to get more details, Source=Microsoft.DataTransfer.ClientLibrary, Type=System.Data.SqlClient.SqlException, Message=Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1). The statement has been terminated, Source=.Net SqlClient Data

How helpful or unhelpful was this error message?
★★★★★

Pipeline status Failed View debug run consumption
Monitor in Azure Metrics Export to CSV

Activity	Status	Duration	Integration runtime	User
Copy file data to...	Failed	16s	AutoResolveIntegration	
if Customer csv file availalbe	Failed	17s		
Check if Customer CSV file ex...	Succeeded	3s	AutoResolveIntegration	

The screenshot displays a data integration pipeline with the following components and status:

- Copy Cust csv file data to the Customer Table**: Failed. Error details: `Microsoft.DataTransfer.Common.Shared.HybridDeliveryException, Message=A database operation failed. Please search error to get more details., Source=Microsoft.DataTransfer.ClientLibrary, Type=System.Data.SqlClient.SqlException, Message=Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1). The statement has been terminated., Source=.Net SqlClient Data Provider, SqlErrorNumber=2627, Class=14, ErrorCode=-2146232060, State=1, Errors=[{Class=14, Number=2627, State=1, Message=Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1)., {Class=0, Number=3621, State=0, Message=The statement has been terminated.,}],'`
- Check if Customer_CSV file exists**: Succeeded.
- If Customer CSV Available**: Failed.

The pipeline status is **Failed**. The error message is: **Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1).**

Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. Failure happened on 'Sink' side. ErrorCode=SqlOperationFailed,Type=Microsoft.DataTransfer.Common.Shared.HybridDeliveryException,Message=A database operation failed. Please search error to get more details.,Source=Microsoft.DataTransfer.ClientLibrary,'Type=System.Data.SqlClient.SqlException,Message=Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1). The statement has been terminated.,Source=.Net SqlClient Data Provider,SqlErrorNumber=2627,Class=14,ErrorCode=-2146232060,State=1,Errors=[{Class=14,Number=2627,State=1,Message=Violation of PRIMARY KEY constraint 'PK_Customer_CustomerID'. Cannot insert duplicate key in object 'SalesLT.Customer'. The duplicate key value is (1).,},{Class=0,Number=3621,State=0,Message=The statement has been terminated.,}],'

The screenshot shows an 'Error details' dialog box for a failed pipeline run. The error message is: 'Cannot access destination table '[SalesLt].[Customer_Table_Ass_7_ques_3]''. The source is 'System.Data.SqlClient.SqlException' with the message 'Invalid object name 'SalesLt.Customer_Table_Ass_7_ques_3''. Below the error details is a feedback section asking 'How helpful or unhelpful was this error message?' with five stars. At the bottom, a table shows the pipeline run history.

Activity Name	Status	Reason	Activity Type	Run Date
Copy file dat...	Failed	Copy data	Copy data	2/13/2024,
if Customer csv file availalbe	Failed	If Condition	If Condition	2/13/2024,
Check if Customer CSV file ex...	Succeeded	Get Metadata	Get Metadata	2/13/2024,

Note: Some times copy fails while trying to copy file cannot be copied from ADLS product.txt to SQL product DB.

This is because of Primary key constraints in the SQL DB,

Because Customer Table already exists, so if you try to insert same duplicate value, Primary key throws error becauseu of duplication

Soltuion: Create a new Table name Customer copy_table or Truncate existing record OR upsert

You can refer assignemnt 6 question 3 solution

I used the approach of Upsert (Update + Insert) Based on one of the column field, I choose CustomerID

Alternatively, you can create a new SQL DB table and insert /copy your records

```
CREATE TABLE TsInfo ( EmpId INT IDENTITY  
PRIMARY KEY, EmpName NVARCHAR(128) NOT  
NULL, Email NVARCHAR(256) )
```

[How To Create Table In Azure SQL Database - Azure Lessons](#)

On the Query window, write the query below.

```
CREATE TABLE Student  
(  
    StudentId INT IDENTITY PRIMARY KEY,  
    FirstName NVARCHAR(128) NOT NULL,  
    LastName NVARCHAR(128) NOT NULL,  
    DateOfBirth DATE NOT NULL  
)
```

« ✓ Validate ▶ Debug ⚡ Add trigger

Get Metadata
Check if Customer CSV file...

If Condition
If Customer CSV Available

True
Copy Cust csv file...

False

General Source **Sink** Mapping Settings User properties

Sink dataset *
SQL_Customer_Table_Dataset Open + New Learn more

Write behavior
☐ Insert ☒ Upsert ☐ Stored procedure

Use TempDB ☒

Key columns
+ New Delete Refresh

☐ Column name

☐ 123 CustomerID

Bulk insert table lock ☐ Yes ☒ No

Table option
☒ None ☐ Auto create table

*Upsert (Update + Insert)
Cust ID*

ADLS_Generic_CSV_DS

Assignment_6_Fore...

Ass_7_ques_3

ADLS_Customer_CSV...

SQL_Customer_Table...

Ass_7_S

Validate

Cancel options

Add trigger

Get Metadata

Check if Customer CSV file exist

If Condition

if Customer csv file availalbe

True

Copy file data to...

False

Parameters

Variables

Settings

Output

Pipeline run ID: 119be205-83bb-4bbf-8c63-17908efe0174

Pipeline status In progress

All status

List

Monitor in Azure Metrics

Export to CSV

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Copy file data to Customer ta...	Queued	Copy data	2/13/2024, 11:58:19 PM	Less than 1s	
if Customer csv file availalbe	In progress	If Condition	2/13/2024, 11:58:18 PM	Less than 1s	
Check if Customer CSV file ex...	Succeeded	Get Metadata	2/13/2024, 11:58:13 PM	4s	AutoResolveIntegration

Validate

Debug

Add trigger

Get Metadata

Check if Customer CSV file...

If Condition

If Customer CSV Available

True

Copy Cust csv file...

False

Parameters

Variables

Settings

Output

Pipeline run ID: ee884d33-588a-4c8d-b22b-96d869718c14

Pipeline status Succeeded

All status

List

View debug run consumption

Monitor in Azure Metrics

Export to CSV

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
Copy Cust csv file data to the...	Succeeded	Copy data	2/14/2024, 12:18:35 AM	15s	AutoResolveIntegration		52d272f8-3e5f-4829-ad98-126cb26f
If Customer CSV Available	Succeeded	If Condition	2/14/2024, 12:18:34 AM	17s			d1266874-bf5a-4d8d-a26f-74152b2f
Check if Customer_CSV file ex...	Succeeded	Get Metadata	2/14/2024, 12:18:31 AM	3s	AutoResolveIntegration		dee3a171-2e73-4066-a500-a8644b0f

Output



Copy to clipboard

[Learn more on output details](#)

```
{
  "dataRead": 217984,
  "dataWritten": 278379,
  "filesRead": 1,
  "sourcePeakConnections": 1,
  "sinkPeakConnections": 2,
  "rowsRead": 847,
  "rowsCopied": 847,
  "copyDuration": 13,
```

Copy Cust csv file data to the... ✓ Succeeded

Details [Refresh](#)

[Learn more on copy performance details from here.](#)

Activity run id: 52d272f8-3e5f-4829-ad98-126cb26b8031



Azure Data Lake Storage Gen2
Region: East US

Succeeded



Azure SQL Database
Region: East US

Data read: ① 217.984 KB
Files read: ① 1
Rows read: 847
Peak connections: ① 1

Interim data written: ① 278.379 KB
Interim rows written: ① 847
Data written: ① 278.379 KB
Rows written: ① 847
Peak connections: ① 2

Copy duration 00:00:13
Throughput: ① 54.496 KB/s

▼ Azure Data Lake Storage Gen2 → Azure SQL Database

Start time 2/14/2024, 12:18:35 AM
Used DIUs ① 4
Used parallel copies ① 1

▼ Duration 00:00:13

Details	Working duration	Total duration
✓ Queue ①		00:00:07
✓ Transfer ①	Listing source ① 00:00:00 Reading from source ① 00:00:00 Writing to interim ① 00:00:00 Writing to sink ① 00:00:00	00:00:04

Data consistency verification ① Not verified



How satisfied or dissatisfied are you with the performance of this copy activity?



5. Create a pipeline name 'Foreach_Example2' to copy the below

information using just one copy activity.

- All the product table data where productid > 100
- All the customer table data where Customer id > 100 and Customer id < 1000

✕

✓ Validate

▶ Debug

⚡ Add trigger

ForEach

ForEach array item
Product Customer

Activities

+

🗑

</>

📄

➡

General

Settings

Activities (0)

User properties

Name *

ForEach array item Product Customer

[Learn more](#)

Description

All the product table data where
productid > 100
- All the customer table data where

Activity state ⓘ

☒ Activated ☐ Deactivated

Search factory and documentation

DEFAULT DIRECTORY

by to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Customer_CSV... SQL_Customer_Table... NEW_Meta_data_test SQL_DB_Generic_DS

<< Validate Debug Add trigger

ForEach

ForEach array item
Product Customer

Activities

Copy data1

</>

General Settings Activities (1) User properties

Sequential ☐

Batch count ^①

Items @createArray('Customer','Product')

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

@createArray('Customer','Product')

Clear contents

Activity outputs Parameters System variables Functions Variables

Copy data1
Copy data1 activity output

Copy data1
Copy data1 pipeline return value

OK Cancel

< Validate Debug Add trigger

ForEach

ForEach array item
Product Customer

Activities

Copy data1

</>

General Settings Activities (1) User properties

Sequential ☐

Batch count ^①

Items @createArray('Customer','Product')

DLSCustomer_CSV...SQL_Customer_Table...NEW_Meta_data_testSQL_DB_Generic_DSAss_7_ques_5_Fore...Ass_6_3_Fon

ValidateDebugAdd trigger

ForEach

ForEach array itemProduct Customer

Activities

Copy data1

GeneralSettingsActivities (1)User properties

Case	Activity
ForEach	Copy data1 1 Activity

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

Ass_7_ques_5_Foreach_Example2 > ForEach array item Product Customer

Copy data

 Copy data1

 copy the below information using just one copy activity.
- All the product table data where productid > 100
- All the customer table data where Customer id > 100 and Customer id < 1000

General Source¹ Sink¹ Mapping Settings User properties

Name * Copy data1 [Learn more](#)

Description
productid > 100
- All the customer table data where Customer id > 100 and Customer id < 1000

Activity state ① ☒ Activated ☐ Deactivated

Timeout ① 0.12:00:00

Retry ① 0

Retry interval (sec) ① 30

ques_3 ADLS_Customer_CSV... SQL_Customer_Table... NEW_Meta_data_test SQL_DB_Generic_DS Ass_7_ques_5_Fore...

Activities

for

Move and transform

Copy data

Data flow

Iteration & conditionals

ForEach

ForEach

ForEach array item
Product Customer

Activities

Copy data1

General Source Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS Open + New Preview data Learn more

Dataset properties

Name	Value
schemaName	SalesLt
tableName	@item()

Use query ☒ Table ☐ Query ☐ Stored procedure

Query timeout (minutes) 120

Isolation level Select...

Partition option ☒ None ☐ Physical partitions of table ☐ Dynamic range

Search factory and documentation

ng way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

SQL_DB_Generic_DS SQLDB_DS Foreach_Example2 x Foreach_Example_2

Validate Debug Add trigger

ForEach

For each table

Activities

Copy table data

General Source Sink Mapping Settings User properties

Source dataset * SQL_DB_Generic_DS Open + New

Dataset properties

Name	Value
schemaName	saleslt
tableName	@item()

Use query ☐ Table ☒ Query ☐ Stored procedure

Query @if(equals(item(),'Customer'),concat('Select * from saleslt.',item()),' where Customerid > 100 and Customerid < 1000'),concat('Select * from saleslt.',item()),' where Productid >100'))

Query timeout (minutes) 120

Isolation level Select...

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@if(equals(item(),'Customer'),concat('Select * from saleslt.',item()),' where Customerid > 100 and Customerid < 1000'),concat('Select * from saleslt.',item()),' where Productid >100'))
```

Clear contents

ForEach iterator Parameters System variables Functions Variables

Search

For each table
Current item

OK Cancel

Transform

Conditionals

ForEach

ForEach array item
Product Customer

Activities

Copy data1

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open

New

Learn more

Dataset properties

Name	Value
folderName_param	@concat('Ass_7_ques_foreach_',item...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

+ New

Quote all text

☒

ADLS_Customer_CSV...

SQL_Customer_Table...

NEW_Meta_data_test

SQL_DB_Generic_DS

Validate

Debug

Add trigger

Transform

Conditionals

ForEach

ForEach array item
Product Customer

Activities

Copy data1

General

Source

Sink

Mapping

Settings

User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open

Dataset properties

Name	Value
folderName_param	@concat('Ass_7_ques_foreach_',item...

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

+ New

Quote all text

☒

Add dynamic content below using any combination of expressions, functions and operators

@concat('Ass_7_ques_5_foreach_',item())

Clear contents

ForEach iterator

Parameters

System variables

Functions

Search

ForEach array item Product Customer

Current item

OK

Cancel

, a brand new and exciting way to build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

Activities

for

Move and transform

Copy data

Data flow

Iteration & conditionals

ForEach

Validate Debug Add trigger

ForEach

ForEach array item Product Customer

Activities

Copy data1

General Source Sink Mapping Settings User properties

Source dataset *

SQL_DB_Generic_DS

Open + New Preview data Learn more

Dataset properties

Name Value

schemaName SalesLt

tableName @item()

Use query

Table Query Stored procedure

Query @if(equals(item(),'Customer').concat(...)

Query timeout (minutes) 120

Isolation level Select...

Search factory and documentation

analytics. Click [here](#) to get started with Fabric Data Factory!

SQL_Customer_Table... NEW_Meta_data_test SQL_DB_Generic_DS

Debug Add trigger

ForEach

ForEach array item Product Customer

Activities

Copy data1

Source Sink Mapping Settings User properties

SQL_DB_Generic_DS

Open

Dataset properties

Name Value

schemaName SalesLt

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#)

```
@if(equals(item(),'Customer').concat('Select * from saleslt.',  
item(),' where Customerid > 100 and Customerid < 1000',  
concat('Select * from saleslt.',item(),' where Productid  
>100'))
```

Clear contents

ForEach iterator Parameters System variables Functions Variables

Search

ForEach array item Product Customer

Current item

Search factory and documentation

Build cloud-first data analytics. Click [here](#) to get started with Fabric Data Factory!

ner_CSV...SQL_Customer_Table...NEW_Meta_data_testSQL_DB_Generic_DS

✓ Validate▶ Debug⚡ Add trigger

data1
🗑️</>

GeneralSourceSinkMappingSettingsUser properties

Sink dataset *

ADLS_Generic_CSV_DSOpen

Dataset properties ⓘ

Name	Value
folderName_param	@item()

Copy behavior ⓘ

Select...

Max concurrent connections ⓘ

Block size (MB) ⓘ

Metadata ⓘ

+ New

Quote all text

✓

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#)

@concat('Ass_7_ques_5_foreach_exam_',item())

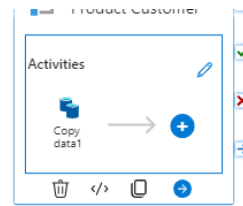
Clear contents

ForEach iteratorParametersSystem variablesFunctions

🔍 Search

ForEach array item Product Customer
Current item

OKCancel



General Source **Sink** Mapping Settings User properties

Sink dataset *

ADLS_Generic_CSV_DS

Open + New [Learn more](#)

Dataset properties

Name	Value
folderName_param	@concat('Ass_7_ques_5_foreach_',item())

Copy behavior

Select...

Max concurrent connections

Block size (MB)

Metadata

+ New

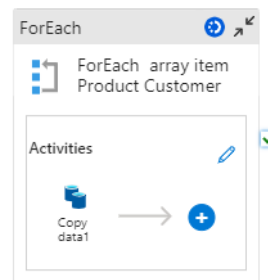
Quote all text

☒

✓ Validate

⊘ Cancel options

⚡ Add trigger



Parameters

Variables

Settings

Output

Pipeline run ID: d2ca7ae1-0602-4347-a6b9-8126a8741a68

Pipeline status

All status ▾ List ▾

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integr.
Copy data1	⌚ Queued	Copy data	2/14/2024, 1:19:26 AM	Less than 1s	
Copy data1	⌚ Queued	Copy data	2/14/2024, 1:19:26 AM	Less than 1s	
ForEach array item Product C...	⌚ In progress	ForEach	2/14/2024, 1:19:26 AM	2s	

Parameters

Variables

Settings

Output

Pipeline run ID: d2ca7ae1-0602-4347-a6b9-8126a8741a68

Pipeline status ✔ Succeeded

All status ▾ List ▾

Showing 1 - 3 of 3 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Copy data1	✔ Succeeded	Copy data	2/14/2024, 1:19:26 AM	1m 15s	AutoResolveIntegratic
Copy data1	✔ Succeeded	Copy data	2/14/2024, 1:19:26 AM	1m 13s	AutoResolveIntegratic
ForEach array item Product C...	✔ Succeeded	ForEach	2/14/2024, 1:19:26 AM	1m 17s	

	Name	Modified	Access tier	Archive status	Blob type	Size
solve problems						
l (IAM)	☐ Address					
	☐ Ass_5_3_Customer_Join_Address_CSV					
tokens	☐ Ass_6_4_three_Tables_foreachCustomer					
	☐ Ass_6_4_three_Tables_foreachCustomerAddress					
	☐ Ass_6_4_three_Tables_foreachProduct					
	☐ Ass_7_ques_5_foreach_Customer					
	☐ Ass_7_ques_5_foreach_Product					
	☐ Assignment_5_5_Address_Table_CSV					
	☐ Assignment_5_5_Conn_Product_Table_CSV					

Search resources, services, and docs (G+7)

gopinathm...@ent...
DEFAULT DIRECTORY (GOPINATHM...)

Ass_7_ques_5_foreach_Customer/data_0e69449f-4a20-...
Blob

Save Discard Download Refresh Delete

Overview Versions Edit Generate SAS

1 CustomerID,NameStyle,Title,FirstName,MiddleName,LastName,Suffix,CompanyName,S
2 101,False,"Mr.",,"William",,"J.",,"Conner",,"Urban Sports Emporium",,"adventure-v
3 102,False,"Ms.",,"Linda",,"Leste",,"National Manufacturing",,"adventure-works\j
4 106,False,"Ms.",,"Andrea",,"A.",,"Thomsen",,"West Side Mart",,"adventure-works\j
5 109,False,"Mr.",,"Daniel",,"P.",,"Thompson",,"Travel Sports",,"adventure-works\p
6 110,False,"Ms.",,"Kendra",,"N.",,"Thompson",,"Vintage Sport Boutique",,"adventure
7 111,False,"Mr.",,"Scott",,"A.",,"Colvin",,"Westside Cycle Store",,"adventure-work
8 112,False,"Ms.",,"Elsie",,"L.",,"Lewin",,"Town Industries",,"adventure-works\jill
9 113,False,"Mr.",,"Donald",,"M.",,"Thompson",,"Trendy Department Stores",,"adventu
10 114,False,"Mr.",,"John",,"L.",,"Colon",,"Two Wheels Cycle Store",,"adventure-work
11 115,False,"Mr.",,"George",,"Z.",,"Li",,"Security Racks and Locks Wholesalers",,"e
12 118,False,"Mr.",,"Yale",,"Li",,"Rapid Bikes",,"adventure-works\josé",,"yale0@ac
13 119,False,"Ms.",,"Phyllis",,"A.",,"Thomas",,"Red Bicycle Company",,"adventure-wor
14 120,False,"Mr.",,"Pat",,"Coleman",,"Machines & Cycles Store",,"adventure-works\
15 124,False,"Ms.",,"Yuhong",,"Li",,"Nearby Sporting Goods",,"adventure-works\jae
16 127,False,"Mr.",,"Joseph",,"M.",,"Lique",,"Front Sporting Goods",,"adventure-work

Text Preview

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: landing /
Ass_7_ques_5_foreach_Customer

Search blobs by prefix (case-...)
Show deleted objects

Name
☐ [-]
☒ data_0e69449f-4a20-4886-afc...

Ass_7_ques_5_foreach_Product/data_48b9ca22-fe56-4..
Blob

Save Discard Download Refresh Delete

Overview Versions Edit Generate SAS

1 ProductID,Name,ProductNumber,Color,StandardCost,ListPrice,Size,Weigh
2 680,"HL Road Frame - Black, 58","FR-R92B-58","Black",1059.3100,1431.
3 706,"HL Road Frame - Red, 58","FR-R92R-58","Red",1059.3100,1431.5000
4 707,"Sport-100 Helmet, Red","HL-U509-R","Red",13.0863,34.9900,,35,3
5 708,"Sport-100 Helmet, Black","HL-U509","Black",13.0863,34.9900,,35
6 709,"Mountain Bike Socks, M","SO-B909-M","White",3.3963,9.5000,"M",,
7 710,"Mountain Bike Socks, L","SO-B909-L","White",3.3963,9.5000,"L",,
8 711,"Sport-100 Helmet, Blue","HL-U509-B","Blue",13.0863,34.9900,,35
9 712,"AWC Logo Cap","CA-1098","Multi",6.9223,8.9900,,23,2,2005-07-01
10 713,"Long-Sleeve Logo Jersey, S","LJ-0192-S","Multi",38.4923,49.9900
11 714,"Long-Sleeve Logo Jersey, M","LJ-0192-M","Multi",38.4923,49.9900
12 715,"Long-Sleeve Logo Jersey, L","LJ-0192-L","Multi",38.4923,49.9900
13 716,"Long-Sleeve Logo Jersey, XL","LJ-0192-X","Multi",38.4923,49.9900
14 717,"HL Road Frame - Red, 62","FR-R92R-62","Red",868.6342,1431.5000,
15 718,"HL Road Frame - Red, 44","FR-R92R-44","Red",868.6342,1431.5000,
16 719,"HL Road Frame - Red, 48","FR-R92R-48","Red",868.6342,1431.5000,

Text Preview

Authentication method: Access key (Switch to Microsoft Entra user account)
Location: landing /
Ass_7_ques_5_foreach_Product

Search blobs by prefix (case-...)
Show deleted objects

Name
☐ [-]
☒ data_48b9ca22-fe56-4565-b3...

Details Refresh

[Learn more on copy performance details from here.](#)

Activity run id: 48b9ca22-fe56-4565-b3a1-795c14cbf77f



Azure SQL Database
Region: East US

Succeeded



Azure Data Lake Storage Gen2
Region: East US

Data read: ① 718.773 KB
Rows read: 295
Peak connections: ① 1

Data written: ① 1.358 MB
Files written: ① 1
Rows written: ① 295
Peak connections: ① 1

Copy duration 00:01:12
Throughput: ① 239.591 KB/s

▼ Azure SQL Database → Azure Data Lake Storage Gen2

Start time 2/14/2024, 1:19:27 AM
Used DIUs ① 4
Used parallel copies ① 1
▼ Duration 00:01:12

Details	Working duration	Total duration
✔ Queue ①		00:01:07
✔ Transfer ①	Time to first byte ①	00:00:03
	Reading from source ①	
	Writing to sink ①	

Data consistency verification ① Not verified

Details  Refresh

[Learn more on copy performance details from here.](#)

Activity run id: 0e69449f-4a20-4886-afcf-c1d21a7b3901

**Azure SQL Database**
Region: East US

Succeeded

**Azure Data Lake Storage Gen2**
Region: East US

Data read: ⓘ

123.64 KB

Rows read:

376

Peak connections: ⓘ

1

Data written: ⓘ

96.561 KB

Files written: ⓘ

1

Rows written: ⓘ

376

Peak connections: ⓘ

1

Copy duration

00:01:10

Throughput: ⓘ

61.82 KB/s

▼ Azure SQL Database → Azure Data Lake Storage Gen2

Start time

2/14/2024, 1:19:27 AM

Used DIUs ⓘ

4

Used parallel copies ⓘ

1

▼ Duration

00:01:10

Details	Working duration	Total duration
✔ Queue ⓘ		00:01:06
✔ Transfer ⓘ	Time to first byte ⓘ 00:00:00 Reading from source ⓘ 00:00:00 Writing to sink ⓘ 00:00:00	00:00:02

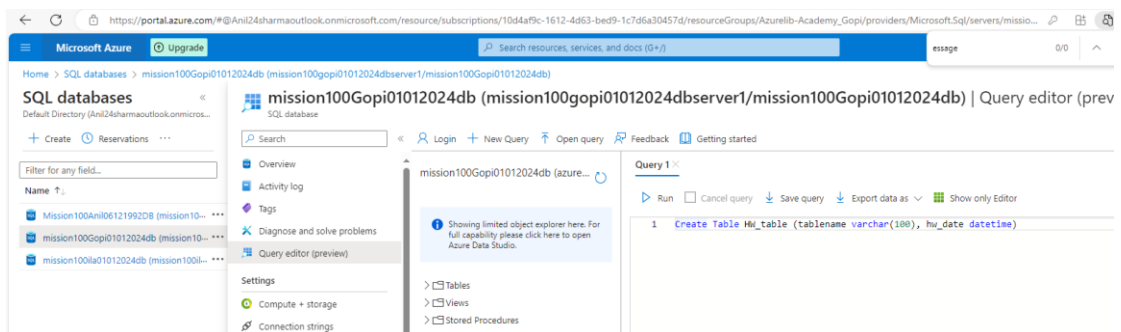
Data consistency verification ⓘ

Not verified

 How satisfied or dissatisfied are you with the performance of this copy activity?

4. Create a pipeline that will create the incremental pipeline using the table instead of HighWaterMark File.

Solution 4:



The screenshot shows the Azure portal interface for the 'mission100Gopi01012024db' database. The 'Query editor (preview)' is open, displaying a SQL query: `Create Table Hbi_table (tablename varchar(100), hu_date datetime)`. The left sidebar shows the 'SQL databases' section with a list of databases, including 'mission100Gopi01012024db'. The top navigation bar includes the 'Upgrade' button and a search bar.

en query  Feedback  Getting started

JRE... 

here. For
open

Query 1

 Run ☐ Cancel query  Save query  Export data as   Show only Editor

```
1 Create Table HW_table (tablename varchar(100), hw_date datetime)
```

< Results Messages

Query succeeded: Affected rows: 0

 Feedback  Getting started

Query 1

 Run ☐ Cancel query  Save query  Export data as   Show only Editor

```
1 Create Table HW_table (tablename varchar(100), hw_date datetime)
2
3 Insert into HW_table values ('Customer' , '1900-01-01 00:00:00')
4
5 Select * from HW_table
```

< Results Messages

 Search to filter items...

tablename	hw_date
Customer	1900-01-01T00:00:00.0000000

ValidateDebugAdd trigger

Lookup

To get HW date

GeneralSettingsUser properties

Name *

To get HW date

Learn more

Description

To get HW date

Activity state ⓘ

☒ Activated☐ Deactivated

Timeout ⓘ

0:12:00:00

Retry ⓘ

0

Properties

GeneralRelated

Name *

Ass_7_4_Table_HighWaterMarkFile

Description

Create a pipeline that will create the incremental pipeline using the table instead of HighWaterMark File

Annotations

+ New

GeneralSettingsUser properties

Source dataset *

SQL_Product_Table_Dataset

Open

New

Preview data

Lee

First row only

☒

Add dynamic content [Alt+Shift+D]

Use query

☐ Table☒ Query☐ Stored procedure

Query *

Select hw_date from HW_table where tablename = 'Customer'

Edit

Query timeout (minutes) ⓘ

120

Isolation level ⓘ

Select...

Partition option ⓘ

☒ None☐ Physical partitions of table ⓘ☐ Dynamic range ⓘ

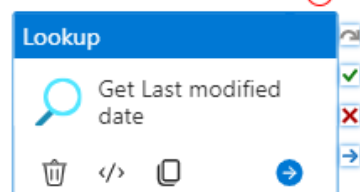
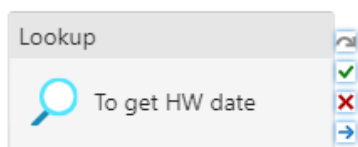
✓ validate ✓ Debug ⚡ Add trigger

Preview data

Linked service: SQL_DB_Linked_service1

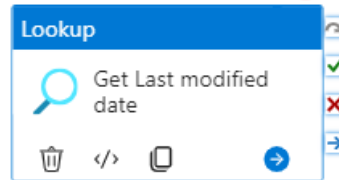
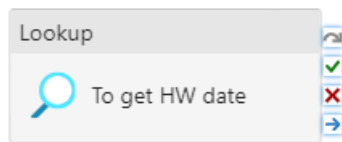
Object:

	hw_date
1	1900-01-01T00:00:00



General Settings User properties

Name *	Get Last modified date	Learn more
Description	Get Last Modified data from SalesLT.Customer	
Activity state ^①	☒ Activated ☐ Deactivated	
Timeout ^①	0.12:00:00	
Retry ^①	0	
Retry interval (sec) ^①	30	



General **Settings** User properties

Source dataset * SQL_Customer_Table_Dataset Open New Preview

First row only ☒

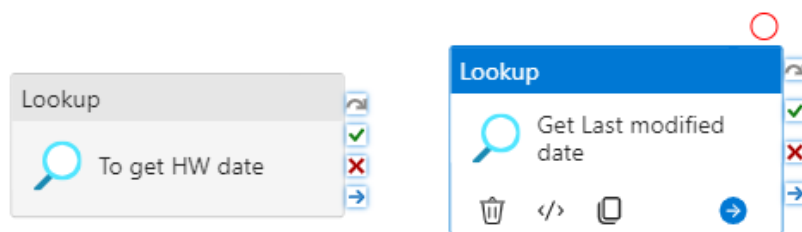
Use query ☐ Table ☒ Query ☐ Stored procedure

Query * SELECT MAX(ModifiedDate) from SALESLT.CUSTOMER Edit

Query timeout (minutes) ① 120

```
8  
9 Step 1: Get the datetime from the HW file. --- Lookup  
0 Step 2: Get the lastmodified datetime from the db table. -- Lookup  
1 Step 3: Copy the records from table where datetime > step 1 datetime  
2         and datetime <= step 2 datetime --> Copy activity  
3 Step 4: update the HW file to step 2 datetime. -->I copy activity  
4
```

```
3  
4 select * from saleslt.customer where modifieddate> 'date from hw' and modifieddate <=  
'last update date from database'
```

General **Settings** User properties

Source dataset * SQL_Customer_Table_Dataset Open New

First row only ☒

Use query ☐ Table ☒ Query ☐ Stored procedure

Query *

```
SELECT MAX(ModifiedDate) AS lastdate  
from  
SALESLT.CUSTOMER
```

Edit

Query timeout (minutes) ①

Isolation level ① Select...

Partition option ① ☒ None ☐ Physical partitions of table ① ☐ Dynamic range ①

i Please preview data to validate the partition settings.

General **Settings** User properties

Source dataset * SQL_Customer_Table_Dataset Open New Preview data

First row only ☒

Use query ☐ Table ☒ Query ☐ Stored procedure

Query *

```
SELECT MAX(ModifiedDate) AS lastdate  
from  
SALESLT.CUSTOMER
```

Edit

Query timeout (minutes) ①

Isolation level ① Select...

Partition option ① ☒ None ☐ Physical partitions of table ① ☐ Dynamic range ①

Preview data

Linked service: SQL_DB_Linked_service1

Object:

	lastdate
1	2009-05-16T16:33:33.123

SQL_Customer_Table...HW_SQL_DB

✓ Validate

✓ Validate copy runtime

▶ Debug

⚙ Add trigger

Lookup

To get HW date

✓

→

Lookup

Get Last modified date

✓

→

Copy data

Copy records from table

✓

→

Copy data

Update the HW file

✓

GeneralSourceSinkMappingSettingsUser properties

Name *

Copy records from table

[Learn more](#)

Description

Copy records from table where record with `modifieddate > HW datetime and modifieddate <= max(modified time)`

Activity state ①

☒ Activated ☐ Deactivated

Timeout ①

0.12:00:00

Retry ①

0

Retry interval (sec) ①

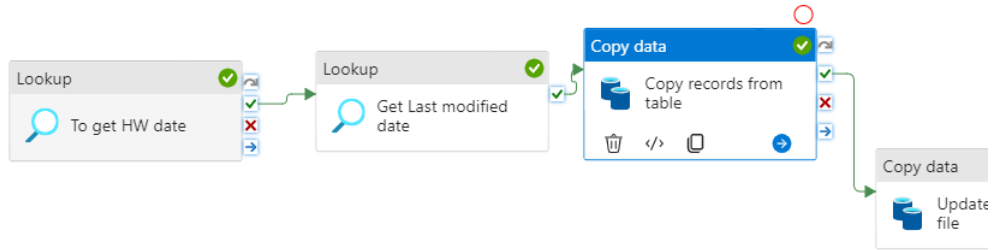
30

Secure output ①

☐

Secure input ①

☐



General **Source** Sink Mapping Settings User properties

Source dataset * SQL_Customer_Table_Dataset Open New Preview data Learn more

Use query Table ☒ Query Stored procedure

Query Select * from Saleslt.customer where ...

Query timeout (minutes) ⓘ 120

Isolation level ⓘ Select...

Partition option ⓘ ☒ None ☐ Physical partitions of table ⓘ ☐ Dynamic range ⓘ

ⓘ Please preview data to validate the partition settings.

Additional columns ⓘ + New

Lookup To get HW date

Lookup Get Last modified date

Copy data Copy records from table

Copy data Update the file

General Source Sink Mapping Settings User properties

Sink dataset * ADLS_Generic_CSV_DS [Open](#) [+ New](#) [Learn more](#)

Dataset properties

Name	Value
folderName_param	Ass_7_4_Table_HighWaterMarkFile

Copy behavior Select...

Max concurrent connections

Block size (MB)

Metadata + New

Quote all text ☒

Lookup To get HW date

Lookup Get Last modified date

Copy data Copy records from table

Copy data Update the HW file

General Source Sink Mapping Settings User properties

Sink dataset * HW_SQL_DB [Open](#) [+ New](#) [Learn more](#)

Write behavior ☐ Insert ☒ Upsert ☐ Stored procedure

Use TempDB ☒

Key columns + New Delete Refresh

Column name

abc tablename

Bulk insert table lock ☐ Yes ☒ No

Table option ☒ None ☐ Auto create table

Lookup

To get HW date

Lookup

Get Last modified date

Copy data

Copy records from table

Copy data

Update the HW file

General

Source

Sink

Mapping

Settings

User properties

Name *

Update the HW file

Learn more

Description

Update the Highwater mark file to Max last modified date

Activity state

☒ Activated ☐ Deactivated

Timeout

0.12:00:00

Retry

0

Retry interval (sec)

30

Secure output

☐

Secure input

☐

Lookup

To get HW date

Lookup

Get Last modified date

Copy data

Copy records from table

Copy data

Update the HW file

General

Source

Sink

Mapping

Settings

User properties

Source dataset *

HW_SQL_DB

Open

New

Preview data

Learn more

Use query

☐ Table ☒ Query ☐ Stored procedure

Query

Select 'Customer' as tablename, @{ac...

Query timeout (minutes)

120

Isolation level

Select...

Partition option

☒ None ☐ Physical partitions of table ☐ Dynamic range

Please preview data to validate the partition settings.

Additional columns

+ New

General Source **Sink** Mapping Settings User properties

Sink dataset * HW_SQL_DB [Open](#) [+ New](#) [Learn more](#)

Write behavior ☐ Insert ☒ Upsert ☐ Stored procedure

Use TempDB ☒

Key columns [+ New](#) [Delete](#) [Refresh](#)

☐ Column name

☒ tablename

Bulk insert table lock ☐ Yes ☒ No

Table option ☒ None ☐ Auto create table

IF `Select 'Customer' as tablename, '@{activity('Get Last modified date').output.firstRow.lastdate}' as hw_date`

Output from above query is : tablename | hw_date

IF same **tablename** and **hwdate** from **source** and Same **tablename** and **hwdate** in **Sink** exist, then instead of inserting, **upsert** I,e update the values in **HW sQL DB based on key column** “

General Source Sink **Mapping** Settings User properties

> Type conversion settings

[Import schemas](#) [Preview source](#) [+ New mapping](#) [Clear](#) [Reset](#) [Delete](#)

☒ Source

Source	Type	Destination	Type
tablename	varchar	tablename	varchar
hw_date	datetime	hw_date	datetime

Home > Storage accounts > gopi01012024db1 | Containers >

landing

Container

Search

<

Upload

+ Add Directory

Refresh

Rename

Delete

Change tier

Acquire lease

Break lease

Give feedback

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Shared access tokens

Manage ACL

Access policy

Properties

Metadata

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / All_7_4_Table_injWaterMarkFile

Search blobs by prefix (case-sensitive)

Show deleted objects

	Name	Modified	Access tier	Archive status	Blob type	Size	Lease state
☐	data_4708de64-3d9f-4a8a-9e8a-ec3b6976f715_a2151c22-9a22-4e46-8767-86e02d99...	2/26/2024, 10:49:30 PM	Hot (inferred)		Block blob	212.88 KiB	Available
☐	data_50abc137-b039-4c5e-80a1-df1b4dd3f5dd_5708fc7f-2feb-4e80-ba5c-4f9f9900...	2/26/2024, 10:20:38 PM	Hot (inferred)		Block blob	212.88 KiB	Available
☐	data_c2a13789-325f-42c7-8335-b6bb98ebca02_e4458625-394a-4e81-bb74-493d0a...	2/26/2024, 10:37:41 PM	Hot (inferred)		Block blob	212.88 KiB	Available
☐	data_e659ac0-4bca-4c4a-a3cc-ca3344474278_df58667e-dc37-4322-b652-29ee97f1...	2/26/2024, 10:36:35 PM	Hot (inferred)		Block blob	212.88 KiB	Available

gopi01012024dbserver1/mission100Gopi01012024db) | Query editor (preview)

Feedback Getting started

Query 1

Run

Cancel query

Save query

Export data as

Show only Editor

```
1 Create Table HW_table (tablename varchar(100), hw_date datetime)
2
3 Insert into HW_table values ('Customer' , '1900-01-01 00:00:00')
4
5 Select * from HW_table
```

Results

Messages

Search to filter items...

tablename	hw_date
Customer	1900-01-01T00:00:00.0000000

```
4
5 Select * from HW_table
```

< Results Messages

Search to filter items...

tablename	hw_date
Customer	2009-05-16T16:33:33.1230000

Final updated value in SQ DB table : "HW_table"

gopio1012024dbserver1/mission100Gopio1012024db | Query editor (preview) ✕

Open query Feedback Getting started

Use...

here. For
to open

Query 1 ✕

Run ☐ Cancel query Save query Export data as Show only Editor

```
1 Create Table HW_table (tablename varchar(100), hw_date datetime)
2
3 Insert into HW_table values ('Customer' , '1900-01-01 00:00:00')
4
5 Select * from HW_table
```

Update



< Results Messages

Search to filter items...

tablename	hw_date
Customer	2009-05-16T16:33:33.1230000